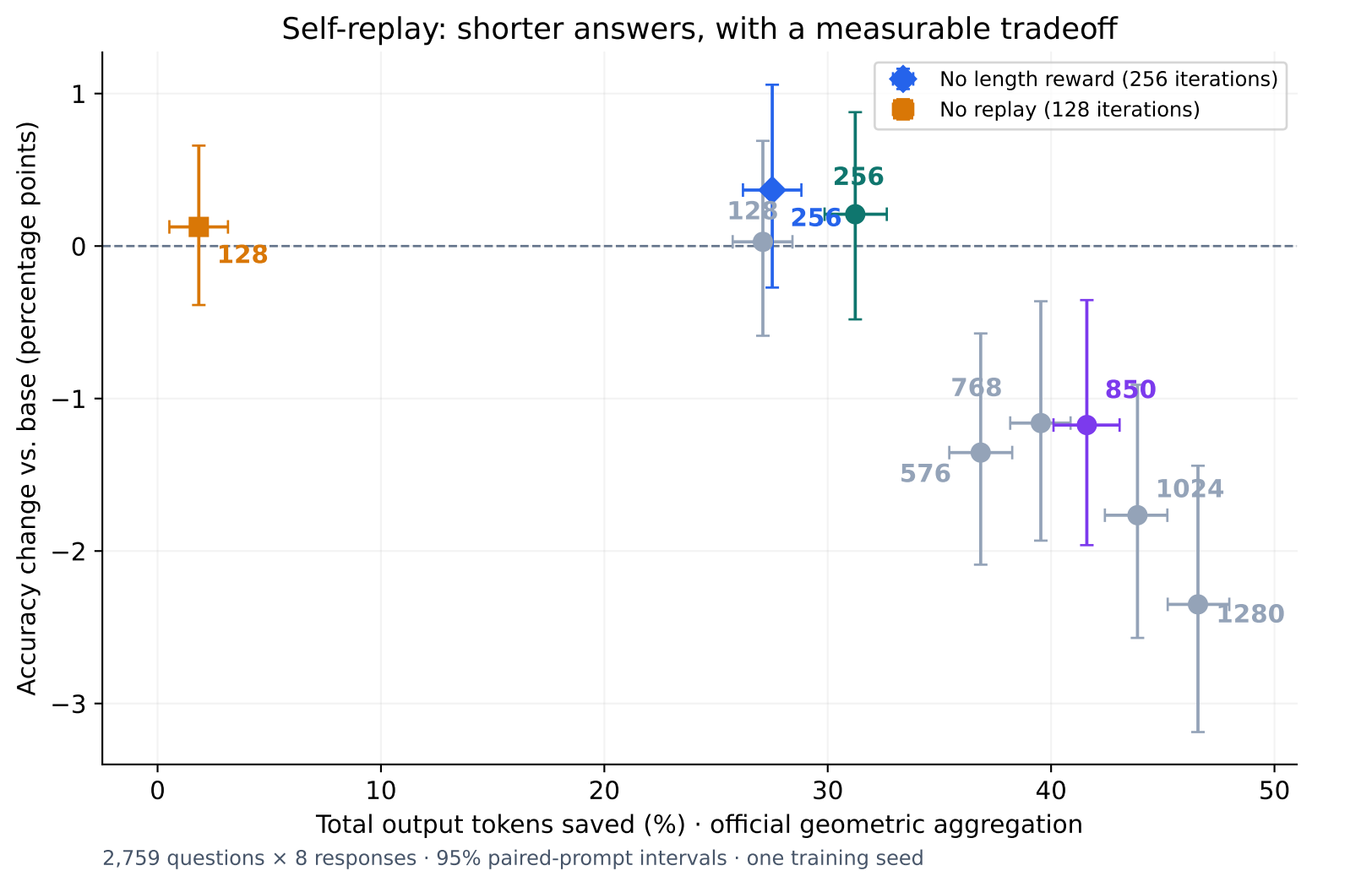
Self-replay: shorter answers, with a measurable tradeoff
No length reward (256 iterations)
No replay (128 iterations)
Accuracy change vs. base (percentage points)
1
0
−1
−2
−3
128
128
256
256
768
576
850
1024
1280
0
10
20
30
40
50
Total output tokens saved (%) · official geometric aggregation
2,759 questions × 8 responses · 95% paired-prompt intervals · one training seed
Checkpoints selected on a repeatedly inspected benchmark; intervals do not include selection or training-seed uncertainty.